

Interconnection Network

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Introduction

An **Interconnection Network (IN)** is a communication infrastructure that connects multiple processors, memory modules, and input/output devices in a parallel computing system, enabling efficient exchange of data and coordination of computations.

In a parallel computer, processors cannot operate independently; they must communicate intermediate results, synchronize execution, and share resources. The interconnection network acts as the communication backbone of the system.

Unlike sequential computers, where a single CPU performs all computations, parallel computers consist of many processing elements that cooperate to solve large computational problems. The efficiency of this cooperation largely depends on the performance of the interconnection network.

Why are Interconnection Networks Needed?

Modern applications such as artificial intelligence, scientific simulations, weather forecasting, cloud computing, and big-data analytics require thousands of processors working simultaneously. Efficient communication between these processors is essential.

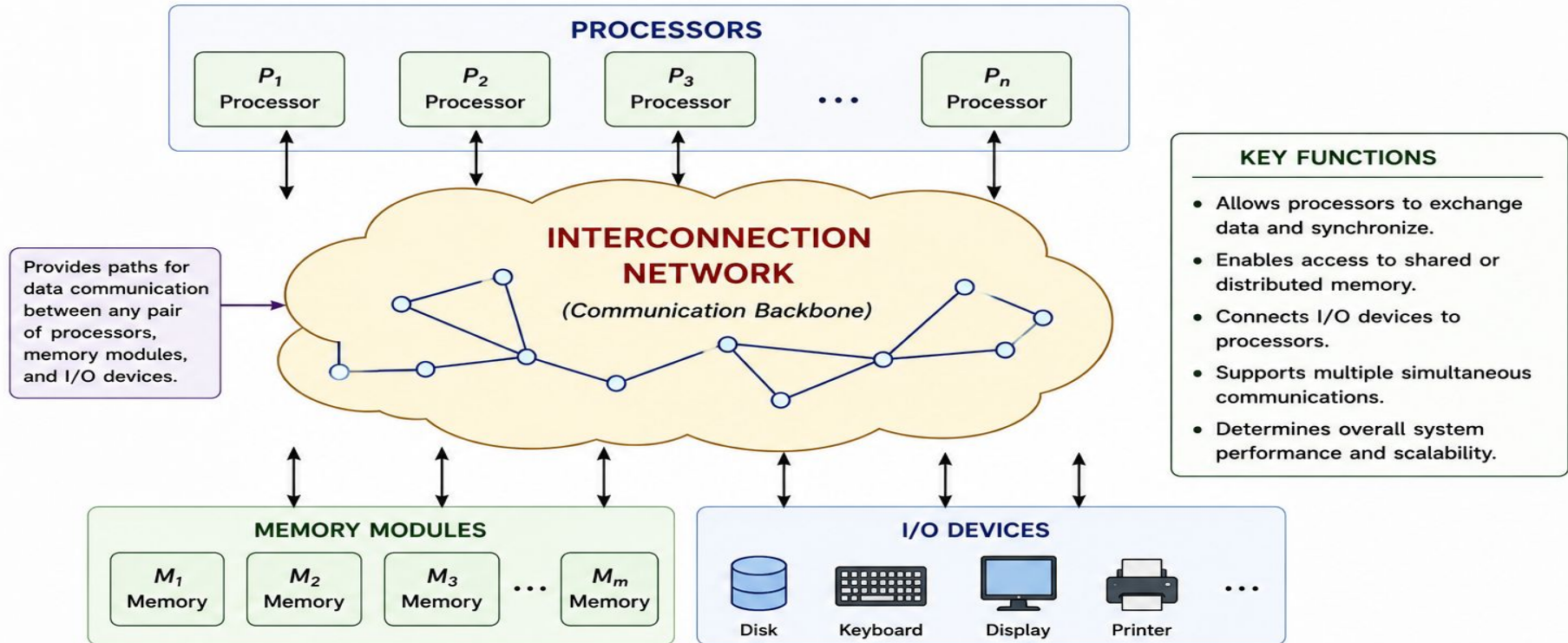
Without an interconnection network:

- Processors cannot exchange information.
- Shared memory cannot be accessed efficiently.
- Parallel algorithms suffer from communication bottlenecks.
- Overall system performance degrades significantly.

Thus, the network is often considered the **nervous system** of a parallel computer.

Role of an Interconnection Network

The interconnection network acts as the communication backbone that enables processors, memory modules and I/O devices to exchange data efficiently.



The interconnection network is the communication backbone that determines the efficiency, speed, and scalability of a parallel computing system.

2. NEED FOR INTERCONNECTION NETWORKS

In a parallel computer, many processors work simultaneously to solve a problem. They must communicate, synchronize and share data and resources. An interconnection network provides the **communication backbone** that makes this cooperation possible.

MAJOR REASONS

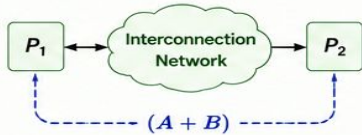
1 Processor Communication

Processors exchange intermediate results required for further computations.

Example:

P_1 computes $(A + B)$.

P_2 needs this result to compute $(A + B) \times C$.

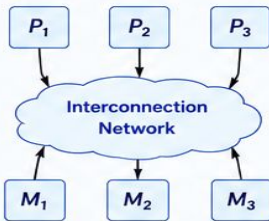


Efficient communication speeds up the overall computation.

2 Shared Memory Access

Multiple processors frequently access common or distributed memory modules.

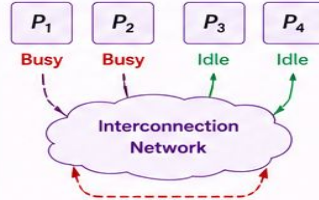
Efficient memory access requires an optimized network.



Ensures fast and consistent access to data.

3 Load Balancing

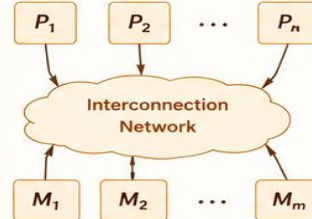
Tasks or data can be transferred from busy processors to idle processors through the network. This improves processor utilization and overall throughput.



Leads to better resource usage and shorter execution time.

4 Scalability

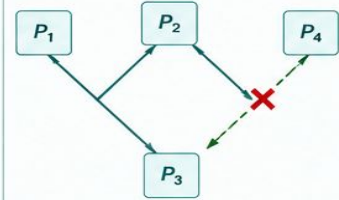
As the number of processors increases, the system must still perform efficiently. A good interconnection network supports scalable growth.



Performance should not degrade significantly with more processors.

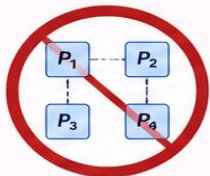
5 Fault Tolerance

If a link or node fails, alternative communication paths in the network allow the system to continue operation.



Improves reliability and system availability.

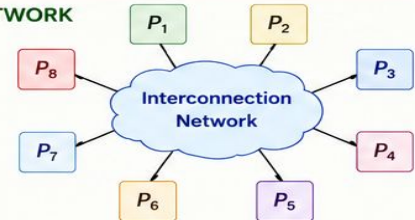
WHAT HAPPENS WITHOUT AN INTERCONNECTION NETWORK?



- ✗ Processors cannot exchange data.
- ✗ Shared memory cannot be accessed efficiently.
- ✗ Load balancing is not possible.
- ✗ Communication bottlenecks occur.
- ✗ Overall system performance degrades.

BENEFITS PROVIDED BY INTERCONNECTION NETWORK

- ✓ Enables fast and efficient communication.
- ✓ Supports high data bandwidth.
- ✓ Improves scalability and system performance.
- ✓ Allows dynamic task distribution and load balancing.
- ✓ Provides reliability and fault tolerance.



Thus, the interconnection network is essential for communication, coordination and cooperation among processors, making it a fundamental requirement for efficient parallel computing.

Evolution of Interconnection Networks

The architecture of communication networks has evolved with increasing processor counts.

Generation

Characteristics

Single Processor

No communication network

Shared Bus

One communication channel

Crossbar Switch

Multiple simultaneous communications

Multistage Networks

Cost-effective communication

Hypercube

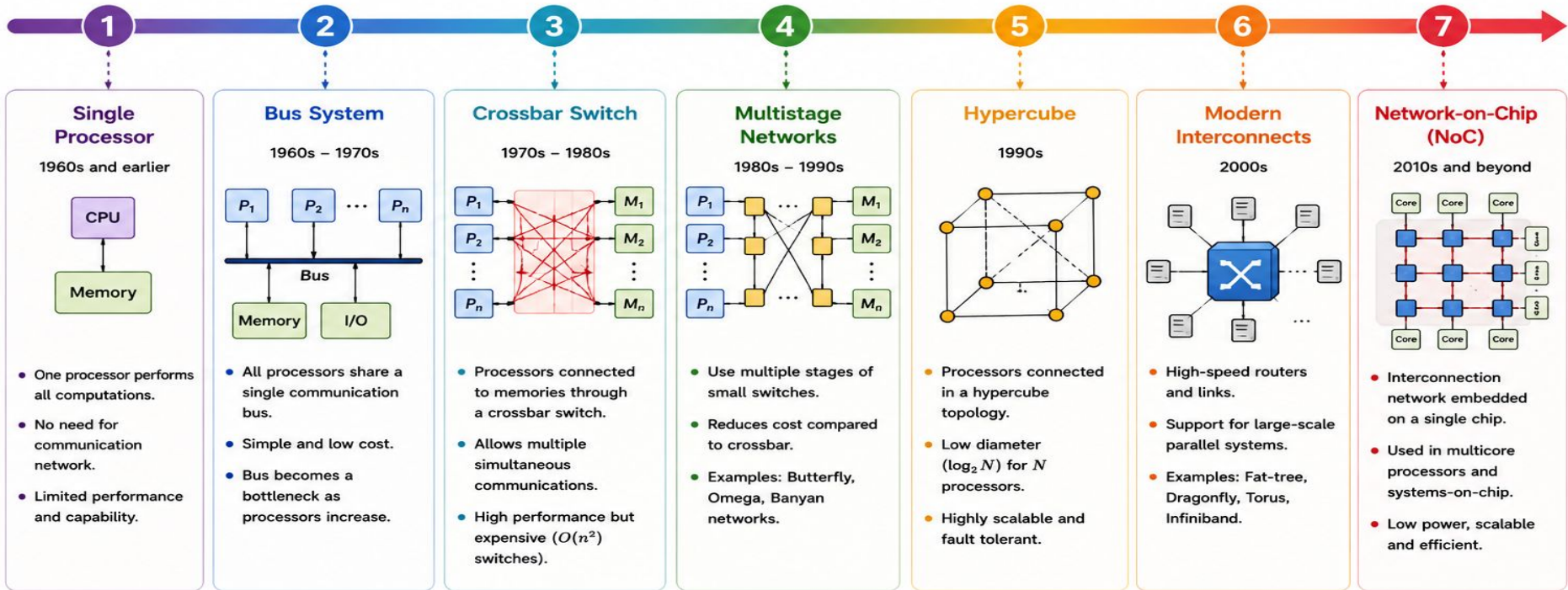
Logarithmic communication distance

Modern NoC

Network-on-Chip architectures

Evolution of Interconnection Networks

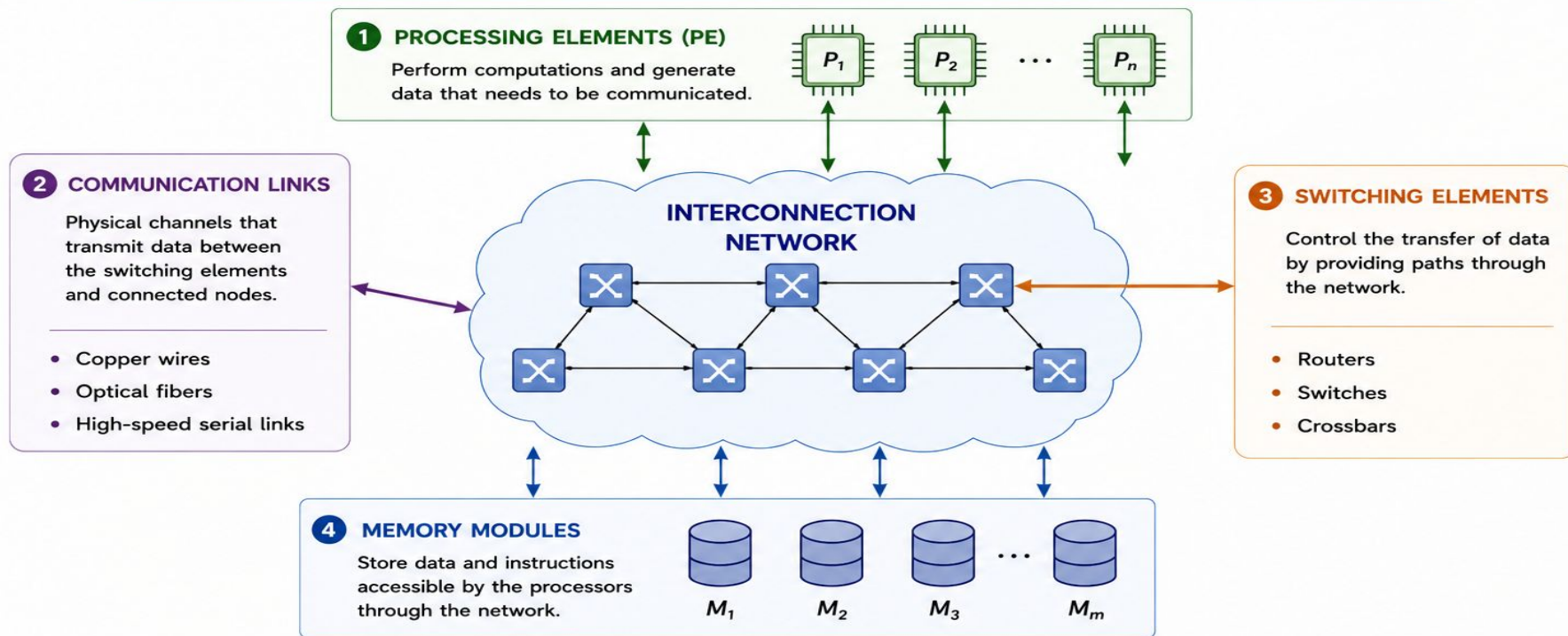
Interconnection networks have evolved from simple shared communication paths to complex, scalable and high-performance architectures.



Summary: From simple buses to advanced NoC architectures, interconnection networks have evolved to meet the demands of higher performance, scalability, and reliability in parallel computing.

Figure 1.4 Components of an Interconnection Network

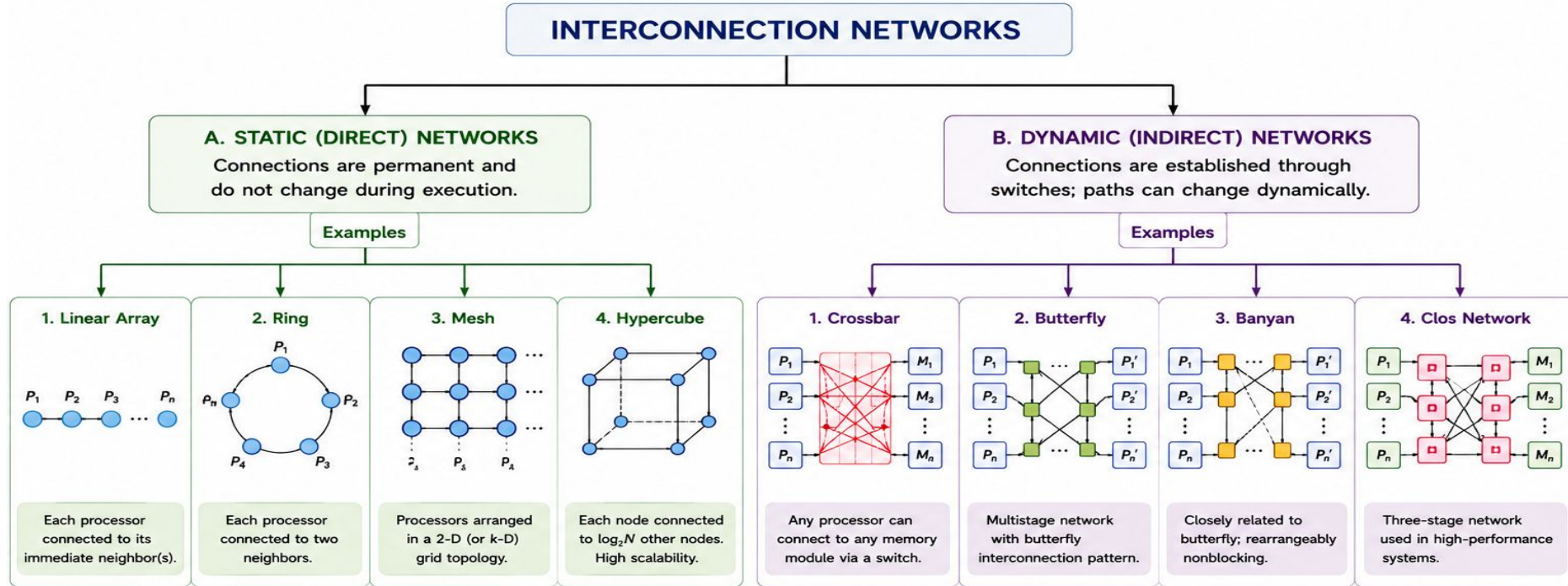
An interconnection network is composed of four main components that work together to enable efficient communication among processors, memory modules and I/O devices.



Summary: These four components—Processing Elements, Communication Links, Switching Elements, and Memory Modules—form the backbone of any interconnection network.

Figure 1.5 Classification of Interconnection Networks

Interconnection networks are broadly classified into two types based on the way processors are connected and how communication paths are established.



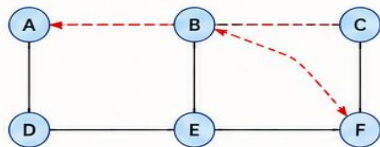
Summary: Static networks have fixed connections and are simple, while dynamic networks use switches to establish paths, offering more flexibility and scalability.

7. PERFORMANCE METRICS OF INTERCONNECTION NETWORK

Performance metrics are used to evaluate and compare interconnection networks.
The efficiency of a network depends on factors such as distance, bandwidth, delay and connectivity.

1. NETWORK DIAMETER

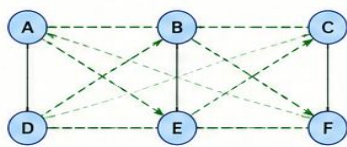
The maximum shortest distance between any two nodes in the network.



Smaller diameter ⇒ lower communication delay

2. AVERAGE DISTANCE

The average of the shortest distances between all pairs of nodes.

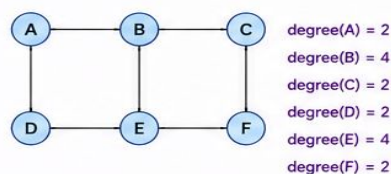


Average of shortest paths
between all node pairs

Smaller average distance ⇒ better efficiency

3. NODE DEGREE

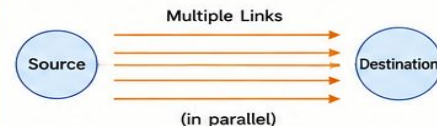
The number of direct links connected to a node.



Higher degree ⇒ more connectivity
but higher cost

4. BANDWIDTH

The total amount of data that can be transferred per unit time.

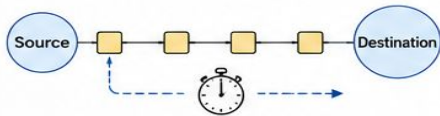


Bandwidth = Number of links × Link speed (bps)

Higher bandwidth ⇒ more data throughput

5. LATENCY (DIELAY)

The time taken for a message to travel from the source to the destination.

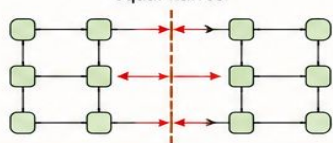


Total latency = Processing delay +
Queuing delay + Transmission delay

Lower latency ⇒ faster communication

6. BISECTION WIDTH

The minimum number of links that must be removed to divide the network into two equal halves.



Bisection Width = Number of links crossing
the bisection

Higher bisection width ⇒ higher throughput

7. THROUGHPUT

The actual rate at which useful data is delivered to the destination.

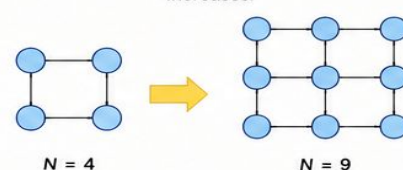


Throughput = Useful data delivered
(bps)

Higher throughput ⇒ better performance

8. SCALABILITY

The ability of the network to maintain performance as the number of nodes increases.



Good scalability ⇒ supports growth

SUMMARY



- These metrics help in designing efficient interconnection networks for parallel computing systems.
- An ideal network should have low diameter, low latency, high bandwidth, high bisection width, high throughput and good scalability.

LEGEND

- Node / Processor
- Switch / Router

Summary

An interconnection network is the communication backbone of a parallel computing system. It enables processors, memory modules, and I/O devices to exchange information efficiently.

The design of the network directly influences communication latency, bandwidth, scalability, and overall system performance.

Understanding the concepts of network topology, routing, switching, and performance metrics provides the foundation for studying advanced architectures such as **Mesh**, **Binary Tree**, **Hypertree**, **Butterfly**, **Hypercube**, and **Shuffle Exchange Networks** covered in the remaining topics of Unit II.

Important Questions

Define an interconnection network.

What is bisection width?

Explain the need for interconnection networks in parallel computing.

Describe the components of an interconnection network with a neat diagram.

Explain the classification of interconnection networks with suitable examples.

Explain the architecture, components, characteristics, and performance metrics of interconnection networks with suitable illustrations. Discuss their role in parallel computing.